Chapter 5.2 — Formula Sheet

Covariance, Correlation & Conditioning Laws

Companion quick-reference to your Notebook 5 syllabus and your *20 Worked Examples* set (Example labels 1–20 cross-referenced below)

A. Covariance Algebra

Definition & Shortcut [Ex 1]

$$\mathrm{Cov}(X,Y) = \mathbb{E}\big[(X-\mu_X)(Y-\mu_Y)\big] = \mathbb{E}[XY] - \mathbb{E}[X]\,\mathbb{E}[Y]$$

Use the shortcut form on a table: only cells with $xy \neq 0$ feed $\mathbb{E}[XY]$. Also $\mathrm{Cov}(X,X) = \mathrm{Var}(X)$.

Bilinearity [Ex 2]

$$\mathrm{Cov}(aX+b,\ cY+d) = ac\,\mathrm{Cov}(X,Y)$$

Shifts (b,d) never matter; only the scale factors do. Covariance is additive in each slot: $\mathrm{Cov}(X+Z,Y) = \mathrm{Cov}(X,Y) + \mathrm{Cov}(Z,Y)$.

Sum / Difference Identity [Ex 3]

$$\mathrm{Var}(X \pm Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) \pm 2\,\mathrm{Cov}(X,Y)$$

Positively correlated legs *add* risk in a sum and *cancel* risk in a difference (pairs trade).

Portfolio Variance [Ex 4–5]

$$\mathrm{Var}(aX+bY) = a^2\mathrm{Var}(X) + b^2\mathrm{Var}(Y) + 2ab\,\mathrm{Cov}(X,Y)$$

For n assets with weights w_i:

$$\mathrm{Var}\Big(\sum_i w_i X_i\Big) = \sum_i w_i^2 \mathrm{Var}(X_i) + 2\sum_{i<j} w_i w_j \mathrm{Cov}(X_i,X_j)$$

Pairs counted once, then doubled — don't forget the 2.

B. Correlation & the Bound

Pearson Correlation [Ex 6]

$$\rho_{X,Y} = \frac{\mathrm{Cov}(X,Y)}{\sigma_X\,\sigma_Y} \in [-1,1]$$

Scale-free: $\rho(aX+b, cY+d) = \mathrm{sgn}(ac)\,\rho(X,Y)$. Always check $\sigma_X, \sigma_Y > 0$ first.

Boundary Cases [Ex 7–8]

$$Y = aX + b \ \Rightarrow\ \rho = \mathrm{sgn}(a) \quad (a \neq 0)$$

$|\rho| = 1$ *iff* Y is an exact linear function of X. The slope's sign alone decides ± 1; its magnitude cancels.

Zero Correlation ≠ Independence [Ex 9]

$$X \perp Y \ \Rightarrow\ \mathrm{Cov}(X,Y) = 0, \qquad \text{but} \quad \mathrm{Cov}(X,Y) = 0 \not\Rightarrow X \perp Y$$

Classic: X symmetric about 0, $Y = X^2$ gives $\mathrm{Cov} = \mathbb{E}[X^3] - 0 = 0$ yet Y is determined by X.

Minimum-Variance Hedge [Ex 10]

$$h^\star = \frac{\mathrm{Cov}(X,Y)}{\mathrm{Var}(Y)}, \qquad \mathrm{Var}(X - h^\star Y) = \mathrm{Var}(X) - \frac{\mathrm{Cov}(X,Y)^2}{\mathrm{Var}(Y)}$$

$$= \mathrm{Var}(X)\,(1-\rho^2)$$

ρ^2 is exactly the fraction of $\mathrm{Var}(X)$ that the hedge removes.

C. Tower Property

Law of Total Expectation [Ex 11–12]

$$\mathbb{E}[X] = \mathbb{E}\big[\mathbb{E}[X \mid Y]\big]$$

Discrete: $\mathbb{E}[X] = \sum_y \mathbb{P}(Y{=}y)\,\mathbb{E}[X \mid Y{=}y]$. Continuous: $\int \mathbb{E}[X \mid Y{=}y]\, f_Y(y)\,dy$. If $\mathbb{E}[X \mid Y] = cY$ then $\mathbb{E}[X] = c\,\mathbb{E}[Y]$ — no integral needed.

Wald's Identity (Random Sums) [Ex 13]

For $S = \sum_{i=1}^{N} X_i$ with iid X_i independent of N:

$$\mathbb{E}[S] = \mathbb{E}[N] \cdot \mathbb{E}[X_i]$$

Reason: $\mathbb{E}[S \mid N{=}n] = n\,\mathbb{E}[X_i]$ is linear in n, then take the outer expectation.

Regime Averaging & Second Moments [Ex 14–15]

$$\mathbb{E}[\text{P\&L}] = \sum_{\text{regimes}} \mathbb{P}(\text{regime}) \cdot \mathbb{E}[\text{P\&L} \mid \text{regime}]$$

$$\mathbb{E}[X^2] = \mathbb{E}\big[\mathbb{E}[X^2 \mid Y]\big], \quad \mathbb{E}[X^2 \mid Y] = \mathrm{Var}(X \mid Y) + \mathbb{E}[X \mid Y]^2$$

D. Eve's Law & Capstones

Law of Total Variance [Ex 16–17]

$$\mathrm{Var}(X) = \underbrace{\mathbb{E}\big[\mathrm{Var}(X \mid Y)\big]}_{\text{within}} + \underbrace{\mathrm{Var}\big(\mathbb{E}[X \mid Y]\big)}_{\text{between}}$$

Check route: $\mathrm{Var}(X) = \mathbb{E}[X^2] - \mathbb{E}[X]^2$ via the Tower Property on X^2 — both routes must agree.

Compound Variance (Random Sum) [Ex 18]

$$\mathrm{Var}(S) = \mathbb{E}[N]\,\mathrm{Var}(X_i) + \mathrm{Var}(N)\,\big(\mathbb{E}[X_i]\big)^2$$

First term: noise inside each fill. Second term: uncertainty in *how many* fills.

Within vs. Between Regime Risk [Ex 19]

When regime means sit far apart relative to within-regime spread, the *between* term dominates $\mathrm{Var}(X)$ — risk is regime risk, not idiosyncratic noise. Compute both pieces and compare.

Law of Total Covariance [Ex 20]

$$\mathrm{Cov}(X,Z) = \mathbb{E}\big[\mathrm{Cov}(X,Z \mid Y)\big] + \mathrm{Cov}\big(\mathbb{E}[X \mid Y], \mathbb{E}[Z \mid Y]\big)$$

Same within/between split as Eve's Law; setting $Z = X$ recovers it exactly. Pipeline: Tower → Eve → Total Covariance.

One-tool-tree reminder: covariance measures linear *co-movement and obeys bilinearity; correlation rescales it into* $[-1,1]$ *and hits the boundary only for exact lines; conditioning on a hidden variable splits every mean, variance and covariance into a* within-group *piece and a* between-group *piece. Tower for the mean, Eve for the variance, Total Covariance for the cross-term — the same split, three times.*